

FINDS: An N-Body Simulator for Computing the Evolution of Large Schools

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Abstract

Animals travelling in cohesive groups yields many collective and individual benefits, such as security against predators and energy efficiency. The replication of such benefits for analogous, biology-inspired technologies like autonomous drone systems requires keen understanding of the cohesion and evolution of systems of these groups, but these investigations are fundamentally limited to small sizes due to the high computational complexity in computing pairwise dynamics at order N^2 . As real schools of fish have been recorded to contain millions of members, FINDS, or Fish INteraction and Dynamics Simulator, was created to efficiently simulate the dynamics of very large schools of fish using common computational techniques employed in N-body simulation. FINDS is implemented as a containerized, modular software suite written predominantly in C (with Python for post-processing) that handles three core simulation processes in order. First, FINDS contains routines for computing the time-derivative of a school, which is a function of the sum of the pairwise interactions for each member of the school. As such, FINDS contains implementations of Brute Force computation, the Barnes-Hut algorithm, and the Fast Multipole Method. Second, FINDS contains custom routines for numerically integrating these systems using embedded Runge-Kutta schemes for adaptive time-stepping. Third, FINDS contains a small library of post-processing modules available to end users to stream-line the analysis of their generated data.

Introduction

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Mathematical Model of Swimmers

FINDS is developed around the far-field source/sink dipole model designed by Mabrouk and Floryan for modelling how passive hydrodynamic interactions at far distances ($r \gg \ell$) effects the evolution of a greater school [1, 2].

Literature Review and Similar Work

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Numerical Methods

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Software Architecture

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Validation

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Performance Evaluation

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Results

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Discussion

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Limitations

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Future Work

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In the development of FINDS, Alvin Ng’s `grav_sim` was used extensively as a reference for implementing a fast N -body simulator in C, particularly in the implementation of the Barnes-Hut algorithm. In addition, the direct advice of Alvin Ng was instrumental to the implementation of linear octrees and parallelization.

Furthermore, generative artificial intelligence (OpenAI’s ChatGPT-5.5 and Google’s Gemini 3.3) were used heavily for initial code generation before being thoroughly tested and

validated to ensure correctness. Artificial intelligence was not used in developing the content of this report, including the abstract, sections, and figures.

Bibliography

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- [1] Mohamed Niged Mabrouk and Daniel Floryan. “Group cohesion and passive dynamics of a pair of inertial swimmers with three-dimensional hydrodynamic interactions”. In: *Bioinspiration and Biomimetics* 20.1 (Nov. 2024), p. 016014. DOI: [10.1088/1748-3190/ad936d](https://doi.org/10.1088/1748-3190/ad936d). URL: <https://doi.org/10.1088/1748-3190/ad936d>.
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Appendix